

Solution Requirements

Date	26 June 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	The system shall provide secure user registration and login using JWT authentication, encrypted passwords, email verification, and password reset functionality.	High
2	Authorization Levels	The system shall support role-based access control for Users and Admins, ensuring only authorized users can access protected modules and financial data.	High
3	External Interfaces	The platform shall integrate with Google Gemini API for AI-powered settlement recommendations and negotiation letter generation, along with SMTP services for email notifications.	High
4	Transactions Processing	The system shall securely process loan details, EMI calculations, financial analysis, settlement predictions, and user data updates in real time.	High
5	Reporting	The system shall generate financial reports, debt analysis reports, settlement summaries, EMI tracking reports, and downloadable analytics dashboards.	Medium
6	Business Rules, etc.	The platform shall apply financial rules such as debt-to-income ratio analysis, EMI burden calculation, overdue loan evaluation, and AI-based settlement recommendation logic.	High
7	Compliance to Laws or Regulations	The system shall follow secure data handling practices, password encryption standards, JWT security mechanisms, and user privacy protection guidelines.	Medium
8	Other	The system shall provide responsive UI design, mobile compatibility, dark mode support, notifications, AI chatbot assistance, and multilingual support.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	TargetMetric/ Acceptance Criteria
1	Performance & Speed	The system shall provide fast page loading, real-time dashboard updates, and quick AI response generation for financial analysis and settlement prediction.	Page loads in less than 2 seconds and API response time under 3 seconds
2	Scalability	The platform shall support increasing numbers of users, loan records, and AI requests without affecting performance.	Supports up to 10,000 concurrent users
3	Security & Data Privacy	The system shall ensure secure authentication, encrypted passwords, protected APIs, secure JWT tokens, and safe handling of user financial data.	Password hashing using bcrypt and HTTPS-secured communication
4	Reliability & Availability	The platform shall provide stable service availability with minimal downtime and proper error handling mechanisms.	99.9% uptime per month
5	Usability & Accessibility	The system shall provide an intuitive, responsive, and user-friendly interface accessible on desktop and mobile devices.	Fully responsive UI and easy navigation across devices
6	Other	The system shall support maintainability, portability, modular architecture, dark mode support, multilingual support, and future scalability for production deployment.	Modular codebase with easy deployment and future enhancements support

